Alejandro Matilla Calvo

alejandromatillacalvo@gmail.com Alejandro Matilla Calvo Greater Boston

Education

Worcester Polytechnic Institute Worcester, MA
B.S. in Robotics Engineering/B.S. in Mechanical Engineering Expected: May 2028

- **4.00/4.00** GPA, Dean's List: Fall 2024, 2025 & Spring 2025, 2026
- Relevant Courses: Thermodynamics, Fluids, Object-Oriented Design, Embedded Systems, Heat Transfer

Projects

CNC Plotter |*Simulation, CAD, Python* Sep. 2025 - May 2026

- Developed CNC plotter that can hold several colors in a barrel and is actuated from a device connected to the ESP's Wi-Fi
- Designed a custom clip compatible with imperial and metric T-slots to use the club's available material
- Worked on the actuation given the coordinates provided by an algorithm

Fruit Harvesting Robot |*Python, VEX, Design* Oct. 2025 - Dec. 2025

- Built VEX robot with custom parts that identifies different fruits, collects, and organizes them
- Designed an elevator with AI camera on the top to keep track of fruits
- Programmed the robot to complete its task 100% autonomously

Nan-011 |*Simulation, CAD, Python, Isaac Sim, RL* June 2026 - Present

- Designed the main shape of the humanoid robot
- Working toward implementing reinforcement learning for autonomous navigation and decision-making
- Designing an electronic architecture small enough to integrate the robot's required capabilities

Experience

Undergraduate Robotics Research |*Robot Design, Assembly, Reinforcement Learning* Aug. 2026 - Present
WPI

- Part of a team developing a robot with adaptable morphology to improve its ability to operate across different environments
- Build and assemble mechanical components while iterating on the robot's physical design
- Contribute to the development and implementation of reinforcement learning methods for robot control and adaptation

Undergraduate ME Research: Pneumatic Actuators |*CAD, EAGLE, Electronic Printing* Sep. 2025 - May 2026
WPI Gateway Labs

- Designed valve for a smart material actuator
- Iterated through mechanical design on Autodesk Fusion to manufacture a part with the highest thermal expansion
- Worked with a ME PhD student to implement the valve and actuate each cell individually

iD Tech Robotics Instructor @ MIT |*VEX Robotics, CAD, Programming, Technical Instruction* June 2026 - Aug 2026
Massachusetts Institute of Technology, Boston

- Taught VEX robotics through robot design, construction, programming, and autonomous and driver-controlled operation
- Guided students through hands-on robotics projects and troubleshooting mechanical and programming challenges
- Adapted instruction to different learning styles and experience levels while supporting students throughout multi-day projects

STEM Tutor |*Mechanics Fundamentals, Analytical Problem-Solving, Technical Communication* Jan. 2023 - June 2025
Madrid, Spain

- Tutored high-school students in core STEM subjects, including Physics, Mathematics, Technical Drawing, and Chemistry
- Guided students through problem-solving methods, emphasizing conceptual understanding and engineering reasoning
- Developed tailored explanations and examples to break down complex mechanics and math concepts clearly

Technical Skills

Mechanical SolidWorks, AutoCAD, Autodesk Fusion, Soldering
Programming Java, C++, Python, Embedded Systems Programming
Languages Spanish (Native), English (Proficient), French (Advanced), Italian (Competent)